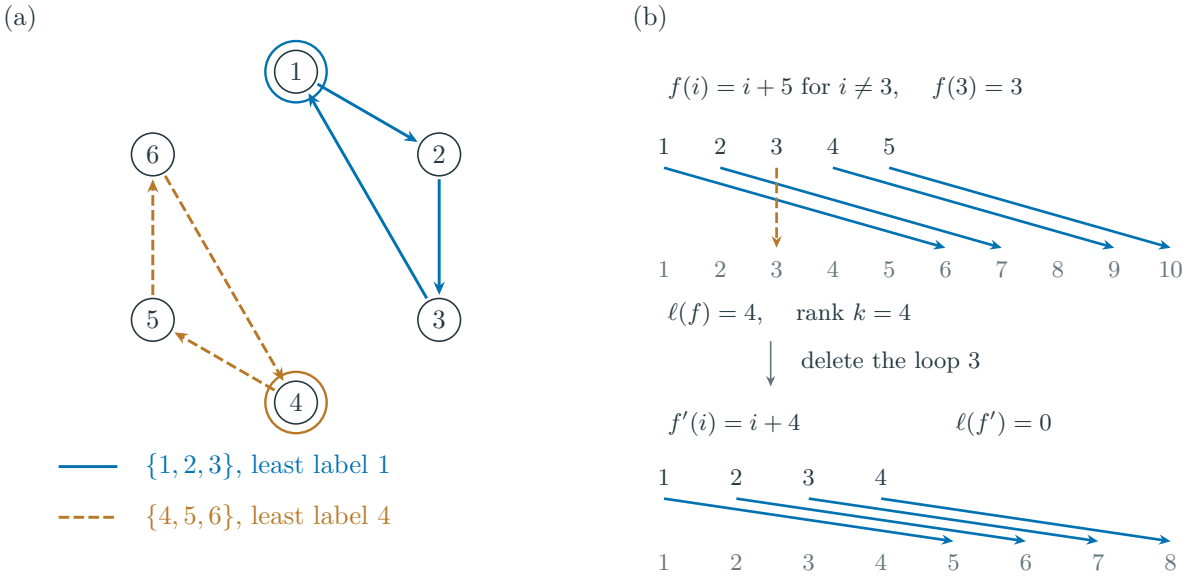

(a)
1
2
3
4
5
6
{1, 2, 3}, least label 1
{4, 5, 6}, least label 4
(b)
$f(i) = i + 5$ for $i \neq 3$, $f(3) = 3$
1 2 3 4 5
1 2 3 4 5 6 7 8 9 10
$\ell(f) = 4$, rank $k = 4$
delete the loop 3
$f'(i) = i + 4$
$\ell(f') = 0$
1 2 3 4
1 2 3 4 5 6 7 8